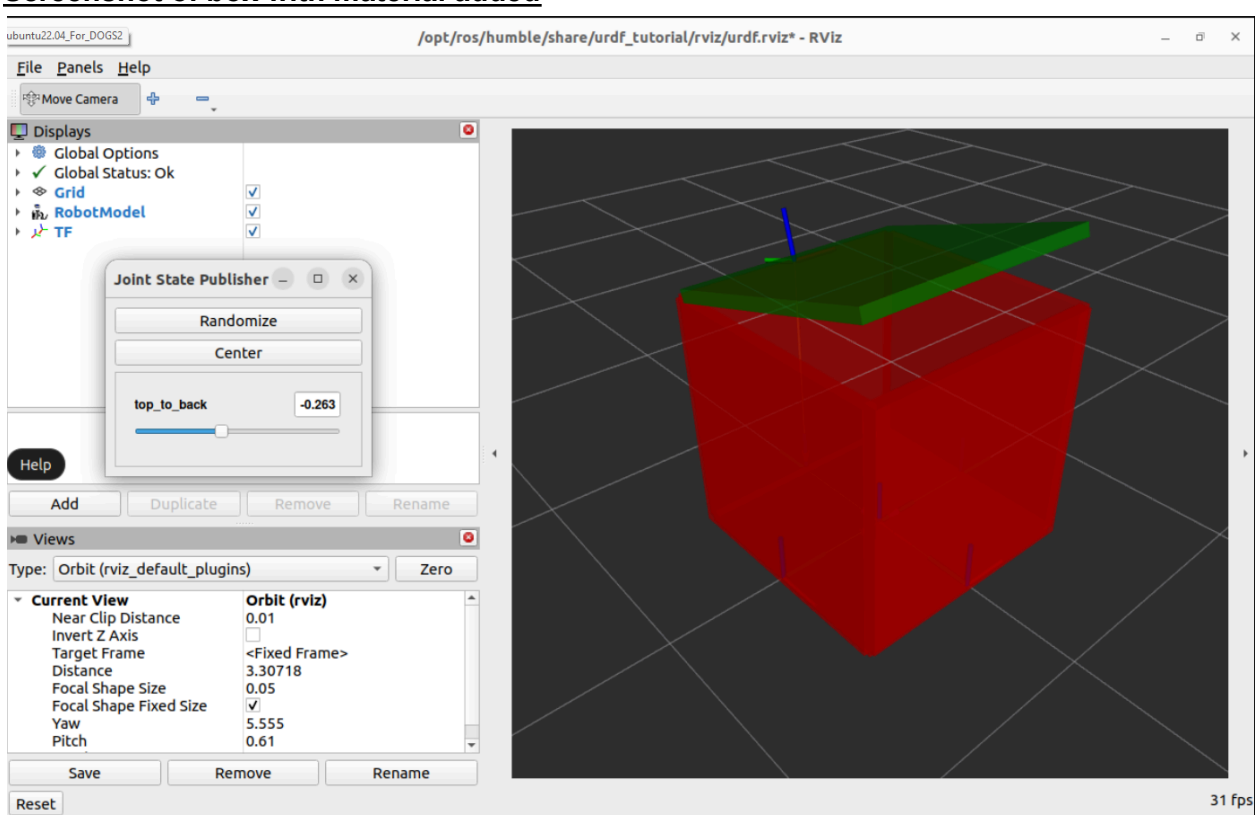
Intro to Robot Programming - Pre lab 6

- Provide a reflection on the assessing and expanding background knowledge sections.
- Add at least one material to the box created in the activity.
- Provide the URDF file and a screenshot in Rviz (with joint state publisher visible) of the completed box with your modification.

The assessing and expanding background knowledge sections provide explanations of how different frames of reference can be mapped onto each other. While it is necessary for there to be an overall coordinate system that these relations exist in, as long as they reside within some uniformly measurable space any number of coordinate systems can be related mathematically. The assessing background knowledge section focused solely on translations, having us practice expressing the same point in different coordinate frames. Then, the expanding background knowledge section lived up to its name by introducing rotations, which could not be handled through simple addition. In order to relate points through coordinate frames that are not oriented in the same way, a rotation matrix has to be applied. This is an nxn square matrix where n = the number of dimensions in the space. This section also reminded us of the right hand rule, which is a helpful tool that can remind us of the positive and negative translations and rotations relative to the 3 cartesian directions.

<u>Screenshot of box with material added</u>